A DISTINGUISHED TRIANGLE RELATING SECTIONS WITH SUPPORT IN A CLOSED SUBSET, GLOBAL SECTIONS AND SECTIONS ON THE OPEN COMPLEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three cross-chapter labels (including a section), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Restriction of a K-injective complex to an open subset is K-injective. Stacks Project, Cohomology of Sheaves, Lemma lemma-restrict-K-injective-to-open (tag 08BS). Allows each of the three derived functors in the triangle to be computed from the same K-injective representative.
- Surjectivity of restriction of sections of an injective module. Stacks Project, Cohomology of Sheaves, Lemma lemma-injective-restriction-surjective (tag 01EA). Gives the final short exact sequence for an injective module.

Source reference. Original section 079V: sources/stacks-chow/tags.txt, source lines 9285–9285. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08BS: sources/stacks-chow/tags.txt, source lines 10577–10577. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01EA: sources/stacks-chow/tags.txt, source lines 1827–1827. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $i : Z \to X$ be the inclusion of a closed subset. Let $U = X \setminus Z$. There is a distinguished triangle*

$$R\Gamma_Z(X, K) \to R\Gamma(X, K) \to R\Gamma(U, K) \to R\Gamma_Z(X, K)[1]$$

in $D(\mathcal{O}_X(X))$ functorial for K in $D(\mathcal{O}_X)$.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ all of whose terms are injective $\mathcal{O}_X$-modules representing K. See Section 079V. Recall that $\mathcal{I}^\bullet|_U$ is a K-injective complex of $\mathcal{O}_U$-modules, see Lemma 08BS. Hence each of the derived functors in the distinguished triangle is gotten by applying the underlying functor to $\mathcal{I}^\bullet$. Hence we find that it suffices to prove that for an injective $\mathcal{O}_X$-module $\mathcal{I}$ we have a short exact sequence

$$0 \to \Gamma_Z(X, \mathcal{I}) \to \Gamma(X, \mathcal{I}) \to \Gamma(U, \mathcal{I}) \to 0$$

This follows from Lemma 01EA and the definitions. □

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